The Looking-Glass Transform: Deriving Quantum Mechanics from Probabilistic Classical Mechanics

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Abstract

We derive non-relativistic quantum mechanics from classical Hamiltonian dynamics via three elementary steps: lifting the equations of motion to a conservation law for the phase-space distribution, Fourier transforming in momentum, and applying a centered-difference approximation to the potential gradient. The resulting equation separates into coupled Schrödinger equations. The classical phase-space distribution is identified with the Wigner quasi-probability function: it satisfies a local conservation law but need not be pointwise non-negative. The Born rule emerges as the momentum marginal. The approximation is exact for quadratic potentials; for anharmonic potentials it differs from the full Wigner–Moyal equation by terms of order $\hbar^2 \partial^3 V / \partial x^3$, while retaining tunneling and interference. We discuss implications for the uncertainty principle and entanglement.

1 Introduction

Quantum mechanics works extraordinarily well, yet its interpretation remains contested. Why does the wavefunction give only probabilities? What happens during measurement? Does the universe branch into many worlds to account for each outcome [1]? These questions persist because the standard formulation takes the Schrödinger equation and the Born rule as postulates, offering no deeper mechanism.

We argue the mechanism is already present in classical Hamiltonian mechanics with probabilistic initial data. The arrow runs from classical to quantum, not the other way: non-relativistic quantum mechanics is revealed *inside* classical phase-space dynamics by a three-step coordinate transformation we call the looking-glass transform. The price of this view is that the phase-space distribution $w(x, p, t)$ is not a classical probability but the

Wigner quasi-probability [2]—a signed measure satisfying a local conservation law whose marginals are genuine probabilities. The Born rule is not a postulate; it is the momentum marginal of w. Measurement is not a collapse; it is a projection onto the observable subspace. There is no measurement problem to solve.

We work in one spatial dimension for clarity; the extension to higher dimensions and multiple particles is straightforward.

2 Quantum Perspective

Consider a tiny one-dimensional non-relativistic particle of mass M in a tiny potential $V(x,t)$. Here the accepted model is the Schrödinger equation, where $|\Psi(x,t)|^2$ gives the probability of observing the particle at location x at time t. The wave function $\Psi(x,t)$ evolves according to the partial differential equation

$$i\hbar\Psi_t = -\frac{\hbar^2}{2M}\Psi_{xx} + V\Psi\,, \tag{1}$$

where $\hbar \approx 1.05 \times 10^{-34}$ joule-seconds.

3 Classical Perspective

Consider a regular one-dimensional non-relativistic particle of mass M in a regular potential $V(x)$. The position of the particle $x(t)$ and momentum of the particle $p(t)$ evolve according to the ordinary differential equation

$$M\frac{dx}{dt} = p\,, \tag{2}$$

$$\frac{dp}{dt} = -\frac{\partial V}{\partial x}\,. \tag{3}$$

4 Three Concepts to Get Quantum Mechanics from Classical Mechanics

What must be added to classical mechanics to get quantum mechanics? We combine the following concepts.

4.1 An ordinary differential equation can be lifted to a linear partial differential equation

Think of an ordinary differential equation initial value problem as evolving a partial differential equation which describes the evolution of the probability distribution of the initial values [3].

ODE:

$$\frac{du_1}{dt} = v_1 \,, \tag{4}$$
$$\frac{du_2}{dt} = v_2 \,, \tag{5}$$
$$\vdots \tag{6}$$
$$\frac{du_n}{dt} = v_n \,. \tag{7}$$

PDE:

$$\frac{\partial w}{\partial t} + \frac{\partial (v_1 w)}{\partial u_1} + \frac{\partial (v_2 w)}{\partial u_2} + \cdots + \frac{\partial (v_n w)}{\partial u_n} = 0 \,. \tag{8}$$

Using the Dirac delta function for the initial value, the PDE recovers the ODE initial value problem: $w(u, 0) = w_0(u) = \delta(u - u_0)$.

4.2 The Fourier transform

$$\hat{\Psi}(\hat{p}) = \int_{-\infty}^{+\infty} dp\, e^{-i\frac{\hat{p}p}{\hbar}} \Psi(p) \,, \tag{9}$$

$$\Psi(p) = \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar}\, e^{+i\frac{\hat{p}p}{\hbar}} \hat{\Psi}(\hat{p}) \,, \tag{10}$$

$$\partial_p \Psi \to i\frac{\hat{p}}{\hbar}\hat{\Psi} \,, \tag{11}$$

$$p\Psi \to i\hbar\partial_{\hat{p}}\hat{\Psi} \,. \tag{12}$$

$$\delta(\hat{p}) = \int_{-\infty}^{+\infty} \frac{dp}{2\pi\hbar}\, e^{i\frac{\hat{p}p}{\hbar}} \,. \tag{13}$$

The use of $\hat{p} = \hbar\omega$ compared to the typical spectral parameter ω greatly compresses the spectrum.

4.3 Centered-difference approximation of $\frac{\partial V}{\partial x}$

From the Taylor expansion of V around x, we have

$$\frac{\partial V}{\partial x} \approx \frac{V(x+\varepsilon/2,t) - V(x-\varepsilon/2,t)}{\varepsilon} + O(\varepsilon^2)\,. \tag{14}$$

We actually approximate $\varepsilon \cdot \frac{\partial V}{\partial x}$, so with first error term:

$$\varepsilon\frac{\partial V}{\partial x} \approx V(x+\varepsilon/2,t) - V(x-\varepsilon/2,t) - \frac{\varepsilon^3}{24}\frac{\partial^3 V(x,t)}{\partial x^3} + O(\varepsilon^5)\,. \tag{15}$$

5 Quantum from Probabilistic Classical Mechanics: The Looking-Glass Transform

We call this the looking-glass transform. It reveals Quantum Mechanics *inside* classical mechanics to a possibly immeasurably good approximation.

We start with the classical Hamiltonian ordinary differential equation:

$$\frac{dx}{dt} = \frac{1}{M}p\,, \tag{16}$$

$$\frac{dp}{dt} = -\frac{\partial V}{\partial x}\,. \tag{17}$$

Using the conservation of probability, this corresponds to the following partial differential equation describing the evolution of the probability density:

$$\frac{\partial w}{\partial t} + \frac{\partial}{\partial x}\left(\frac{1}{M}pw\right) + \frac{\partial}{\partial p}\left(-\frac{\partial V}{\partial x}w\right) = 0\,, \tag{18}$$

or

$$w_t + \frac{p}{M}w_x - \frac{\partial V}{\partial x}w_p = 0\,. \tag{19}$$

Now Fourier transform only with respect to the momentum p and multiply by $i\hbar$. This results in

$$i\hbar\hat{w}_t - \frac{\hbar^2}{M}\hat{w}_{x\hat{p}} + \hat{p}\frac{\partial V}{\partial x}\hat{w} = 0\,. \tag{20}$$

We now change variables. Define

$$z_\pm = x \pm \frac{\hat{p}}{2}\,, \tag{21}$$

So

$$\frac{\partial}{\partial x} \rightarrow \frac{\partial}{\partial z_+} + \frac{\partial}{\partial z_-}\,, \tag{22}$$
$$\frac{\partial}{\partial \hat{p}} \rightarrow \frac{1}{2}\frac{\partial}{\partial z_+} - \frac{1}{2}\frac{\partial}{\partial z_-} \tag{23}$$

This transforms (20) into

$$i\hbar \hat{w}_t - \frac{\hbar^2}{2M}\left[\left(\frac{\partial}{\partial z_+}\right)^2 - \left(\frac{\partial}{\partial z_-}\right)^2\right]\hat{w} + \hat{p}\frac{\partial V}{\partial x}\hat{w} = 0 \tag{24}$$

Next is the only approximation. We use (14) with $\varepsilon = \hat{p}$:

$$\hat{p}\frac{\partial V}{\partial x} \approx V(x + \hat{p}/2, t) - V(x - \hat{p}/2, t) \tag{25}$$

This transforms (20) into

$$i\hbar \hat{w}_t - \frac{\hbar^2}{2M}\left[\left(\frac{\partial}{\partial z_+}\right)^2 - \left(\frac{\partial}{\partial z_-}\right)^2\right]\hat{w} + [V(z_+, t) - V(z_-, t)]\hat{w} = 0\,. \tag{26}$$

While this does not yet look like Schrödinger's equation, this is a linear differential equation, and we can look at separation of variables to construct a general solution to (26). Suppose $\hat{w} = \sum \psi^+(z_+, t)\psi^-(z_-, t)$ where each of these terms satisfy the above, we have

$$\mp i\hbar \psi_t^\pm = -\frac{\hbar^2}{2M}\psi_{zz}^\pm + V(z_\pm, t)\psi^\pm\,. \tag{27}$$

Here z takes the respective roles of z_+ and z_-.

For real potential $V(x,t)$, $\psi^+(z_+, t) = \Psi^*(z_+, t)$ and $\psi^-(z_-, t) = \Psi(z_-, t)$, where $\Psi(x,t)$ is the original Schrödinger wavefunction (1).

If additionally $V(x,t)$ is time-independent, then $\psi^+ = e^{+i\lambda t}\psi_\lambda^*$ and $\psi^- = e^{-i\lambda t}\psi_\lambda$ where the ψ_λ are eigenvectors of the original Schrödinger's equation (1). These form a complete basis for the solution of PDE for $\hat{w}$:

$$\hat{w}(t, z_+, z_-) = \sum_{\lambda,\lambda'} C_{\lambda,\lambda'} e^{i(\lambda-\lambda')t}\psi_\lambda^*(z_+)\psi_{\lambda'}(z_-)\,. \tag{28}$$

where the $C_{\lambda,\lambda'}$ are determined by the initial distribution w_0. Assuming ψ_λ are normalized by $\int_{-\infty}^{+\infty} dz\, |\psi_\lambda(z)|^2 = 1$, we have:

$$C_{\lambda,\lambda'} = \int_{-\infty}^{+\infty} dz_+ \int_{-\infty}^{+\infty} dz_-\, \psi_\lambda(z_+)\psi^*_{\lambda'}(z_-)\hat{w}_0(z_+, z_-)\,, \tag{29}$$

$$= \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} d\hat{p}\, \psi_\lambda(x + \frac{\hat{p}}{2})\psi^*_{\lambda'}(x - \frac{\hat{p}}{2})\hat{w}_0(x, \hat{p})\,. \tag{30}$$

Note C is Hermitian and w is normalized. This implies $C = \sum_k c_k a_k^\dagger a_k$ for real eigenvalues c_k, unit left eigenvectors a_k (so $a_k C = c_k a_k$), and $\sum_k c_k = 1$.

The quantity $w(x, p, t)$ is precisely the Wigner quasi-probability distribution [2], and $\hat{w}$ is its Fourier transform in p. The looking-glass transform has run the standard arrow in reverse: rather than starting from a wavefunction and constructing w, we started from the classical Liouville equation and found that w already encodes a Schrödinger wavefunction. Quantum mechanics was present in classical probabilistic mechanics all along; the transform makes it visible.

Intuition. Quantum mechanics viewed through the lens of the Schrödinger equation carries essentially no classical intuition. Here is a bijection between the two worlds. First, any classical distribution can, to $O(\hbar^2)$, be transformed to a quantum equivalent structure. The reverse takes one additional mental leap: a uniform background probability. $w \to w + c$ for some constant c solves the local probability flux equation (19) just fine. This is the nod to quasi-probability. For many interesting Quantum configurations, the corresponding real quasi-probability,

$$w(x, p, t) = \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar}\, e^{+i\frac{\hat{p}p}{\hbar}}\, \Psi^*(x + \frac{\hat{p}}{2}, t)\Psi(x - \frac{\hat{p}}{2}, t) \tag{31}$$

is real-valued, integrates to 1, but not necessarily non-negative everywhere. If you view this as a shift from a uniform unknowing: a constant probability background c, then the intuition carries. If you prefer no background, then negative w is evidence against that configuration. This might be a way for classical intuition to carry over to the quantum domain.

6 Error Estimate

The centered-difference approximation (14) is exact for linear and quadratic potentials. For other potentials, the leading error term is

$$\hat{p}\frac{\partial V}{\partial x} \approx V(x + \frac{\hat{p}}{2}) - V(x - \frac{\hat{p}}{2}) + \left[-\frac{\hat{p}^3}{24}\frac{\partial^3 V(x)}{\partial x^3}\right]\,. \tag{32}$$

For the approximation to be closer to exact, then the PDE (19) would have to be modified to

$$w_t + \frac{p}{M} w_x - \frac{\partial V}{\partial x} w_p = \left[-\frac{\hbar^2}{24} \frac{\partial^3 V}{\partial x^3} w_{ppp} \right] . \tag{33}$$

This may look like the first quantum correction to classical mechanics, but is actually the first correction to an excellent approximation viewed through the $\hat{p}$ coordinate transformation.

7 The Born Projection

The standard Born rule states that at time t, the probability density of observing position x is $|\Psi(x,t)|^2$. We recover this as the instantaneous marginal of w over momentum, which we call the Born projection:

$$\begin{aligned}
\operatorname{born} w &= \int_{-\infty}^{+\infty} dp \, w(x,p,t) \, , && (34)\\
&= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar} \, e^{i\frac{\hat{p}p}{\hbar}} \, \hat{w} \, , && (35)\\
&= \int_{-\infty}^{+\infty} d\hat{p} \, \hat{w} \int_{-\infty}^{+\infty} \frac{dp}{2\pi\hbar} \, e^{i\frac{\hat{p}p}{\hbar}} \, , && (36)\\
&= \int_{-\infty}^{+\infty} d\hat{p} \, \hat{w} \, \delta(\hat{p}) \, , && (37)\\
&= \hat{w}|_{\hat{p}=0} \, , && (38)\\
&= \sum_{\lambda,\lambda'} C_{\lambda,\lambda'} e^{i(\lambda-\lambda')t} \psi^*_\lambda(x) \psi_{\lambda'}(x) \, , && (39)\\
&= \sum_k c_k \sum_{\lambda,\lambda'} (a_k)^*_\lambda (a_k)_{\lambda'} e^{i(\lambda-\lambda')t} \psi^*_\lambda(x) \psi_{\lambda'}(x) \, , && (40)\\
&= \sum_k c_k \left| \sum_\lambda (a_k)_\lambda e^{-i\lambda t} \psi_\lambda(x) \right|^2 . && (41)
\end{aligned}$$

This is the standard Born rule of quantum mechanics, derived here as the momentum marginal of the phase-space distribution. The dynamics are unaffected—measurement is the projection of w onto the observable subspace, not a physical collapse. The Born rule emerges without decoherence [4] and without additional postulates; it is the momentum marginal of the Wigner quasi-probability.

8 Demystifications

8.1 Heisenberg's Uncertainty Principle

There is nothing wrong with a distribution that specifies the precise initial position and momentum of the particle. However, we can only observe the marginal over momentum, which makes the view fuzzy.

Substitution of $w_0(x,p) = \delta(x-x_0)\delta(p-p_0)$ into the formula for $C_{\lambda,\lambda'}$ gives the cross-Wigner function as the coefficients of the expansion of $\hat{w}$ in the eigenbasis of the Schrödinger equation:

$$C_{\lambda,\lambda'}[\delta(x-x_0)\delta(p-p_0)] = \int_{-\infty}^{+\infty} d\hat{p}\, \psi_\lambda(x_0+\frac{\hat{p}}{2})\psi^*_{\lambda'}(x_0-\frac{\hat{p}}{2})e^{-i\frac{\hat{p}p_0}{\hbar}}\,. \tag{42}$$

For a symmetric potential V, the eigenfunctions ψ_λ are either symmetric or antisymmetric, $\psi_\lambda(-x) = \sigma_\lambda\psi_\lambda(x)$, where $\sigma_\lambda = \pm 1$. We can use this to compute $C_{\lambda,\lambda'}$ for the $x_0 = 0$, $p_0 = 0$ case:

$$\begin{aligned}
C_{\lambda,\lambda'}[\delta(x)\delta(p)] &= \int_{-\infty}^{+\infty} d\hat{p}\, \psi_\lambda(+\frac{\hat{p}}{2})\psi^*_{\lambda'}(-\frac{\hat{p}}{2})\,, && (43)\\
&= \sigma_{\lambda'}\int_{-\infty}^{+\infty} d\hat{p}\, \psi_\lambda(\frac{\hat{p}}{2})\psi^*_{\lambda'}(\frac{\hat{p}}{2})\,, && (44)\\
&= 2\sigma_{\lambda'}\int_{-\infty}^{+\infty} du\, \psi_\lambda(u)\psi^*_{\lambda'}(u)\,, && (45)\\
&= 2\sigma_\lambda\delta_{\lambda,\lambda'}\,. && (46)
\end{aligned}$$

The Born projection yields

$$\mathrm{born}\,\delta(x)\delta(p) = 2\sum_\lambda \sigma_\lambda|\psi_\lambda(x)|^2\,. \tag{47}$$

This is the expansion of $\delta(x)$ as a distribution—what we would expect for a particle at the bottom of the well with no momentum.

8.2 Quantum Entanglement

The above derivation is for a single particle in a single space dimension, but it is trivial to extend to multiple particles in three spatial dimensions. Entanglement amounts to a dependent choice of initial distribution, so $\hat{w}$ cannot be factored into independent components for each particle. The underlying mechanism evolves independently; it is only the initial distribution and the momentum marginal that make it look like there is something mysterious. It is a consequence of dependent distributions and the projection.

For two particles with single-particle eigenstates $\hat{\alpha} = \psi_\lambda$ and $\hat{\beta} = \psi_\mu$, the coefficient matrix C extends to the two-particle eigenbasis $\{|\alpha\alpha\rangle, |\alpha\beta\rangle, |\beta\alpha\rangle, |\beta\beta\rangle\}$. The four maximally entangled Bell states are pure states $C = a\, a^\dagger$ with

$$a_{\Phi^\pm} = \tfrac{1}{\sqrt{2}}(|\alpha\alpha\rangle \pm |\beta\beta\rangle)\,, \tag{48}$$
$$a_{\Psi^\pm} = \tfrac{1}{\sqrt{2}}(|\alpha\beta\rangle \pm |\beta\alpha\rangle)\,. \tag{49}$$

None of these factor as $|s_1\rangle \otimes |s_2\rangle$; entanglement is precisely this non-factorizability of a. The outer product $a\, a^\dagger$ generates off-diagonal coherence terms in $\hat{w}$ that distinguish entangled states from classical mixtures, which would have C diagonal.

8.3 The Wigner Quasi-Probability and Positivity

A useful classical intuition dispels the apparent strangeness of a signed distribution. The Liouville equation (19) is a statement about *flux*, not about absolute values. If w satisfies it, so does $w + c$ for any constant c: a spatially and temporally uniform background has zero gradient and contributes nothing to the flux. Classical Hamiltonian dynamics is entirely insensitive to a constant offset in the phase-space distribution. Quantum mechanics, in this view, describes the evolution of probability *fluctuations* above a locally uniform background; the regions where w dips below zero are simply where the fluctuation is negative relative to the normalization reference level, not where probability is physically absent.

The phase-space distribution $w(x, p, t)$ satisfies a local conservation law and has genuine probability densities as its marginals, but it need not be pointwise non-negative. For any normalized pure state ψ, the Wigner function satisfies the strict bound

$$-\frac{1}{\pi\hbar} \leq w(x, p, t) \leq \frac{1}{\pi\hbar}\,, \tag{50}$$

and negativity in specific phase-space regions has been measured directly via quantum state tomography [**?**]. Hudson's theorem states that $w \geq 0$ everywhere if and only if the state is Gaussian; negativity is precisely the signature of non-classical states such as Fock states or superpositions.

This does not undermine the framework. The Born projection $\mathrm{born}\, w = \int dp\, w = |\Psi(x, t)|^2 \geq 0$ is a genuine probability density regardless of the sign of w, as is the momentum marginal $\int dx\, w = |\tilde{\Psi}(p, t)|^2 \geq 0$. The dynamics are those of a signed measure conserved by Hamiltonian flow, and all observable predictions follow from its marginals.

Tunneling and interference are not lost: they are carried by the retained Schrödinger terms, not by the small Moyal correction terms that are dropped for anharmonic potentials.

8.4 Many Worlds

The many-worlds interpretation [1] posits branching universes to resolve the measurement problem: if the wavefunction never collapses, what selects the observed outcome? In the present framework this question does not arise. The dynamics are deterministic classical Hamiltonian evolution of the phase-space distribution w. The apparent indeterminacy of quantum mechanics arises entirely from the Born projection—the momentum marginal—which discards momentum information that is inaccessible to a position measurement. This is ordinary marginalization of a phase-space distribution, identical in form to how a classical observer recovers a position probability from an (x, p) distribution. One world, definite but partially observable, suffices; the measurement problem is a projection, not a paradox.

9 Further Directions

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References

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